

VOLLEY Gen5 | Final-review evidence handout

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Research question. Can a shared, commandable CubeSat deployer provide useful release states while meeting the installed mass and integration burden of a $12 \times 3U$ reference case?

Decision. The evaluated Gen5 configuration is **not selected** for that case. The computational study records both conditional orbital benefit and adverse system findings. No VOLLEY hardware was built or tested.

Finding	Result and evidence boundary
Release speed	16.029 m/s belongs to a historical periodic-force model. P118's finite 3-D analytic force screen gives 12.448 m/s under ideal phase and omitted circuit losses. It is not an independently verified rating.
Electrical chain	Historical gross draw is 2,782 J; its 124.488 J compact-output remainder resolves to assumed converter, auxiliary and numerical terms. A conditional finite-force full-winding bank rerun gives 2,099 J gross, without selected hardware.
Installed mass	Modeled dry mass is 126.6 kg, or 10.55 kg per 3U customer, against an approximate 6 kg spring-canister comparator. The predeclared 15% parity band fails.
CAD fit	The reference side-fed layout needs 537 mm within 526 mm clear width and overlaps the track by $32,915 \text{ mm}^3$ per cassette. The separate widened R1 geometry clears twelve scripted routes; it lacks a selected feeder, retention and revised mass.
Orbital value	+28.8008 km immediate two-body axis rise is checked for an assumed historical 16.029 m/s release , not as achieved Gen5 behavior. The $1.60 \times$ atmosphere-dependent lifetime figure remains open.
Matched mission	A single assumed target yields ideal host propellant of 2.693 kg spring versus 0.827 kg at the finite Gen5 <i>ideal upper</i> speed. No bounded twelve-shot campaign closes; this is not a proven service advantage.

Evidence classes. The pack contains source scripts, captured run outputs, numerical cross-checks, FreeCAD native review assemblies, STEP exports, figures, reports and an explicit claim-to-source map. These establish only their stated models, geometry and boundary conditions. They do not establish payload qualification, supplier component ratings, host/provider approval, customer demand or flight readiness.

Next gate. Independently check finite integrated thrust; rerun the coupled circuit and trajectory; select a feasible feeder and reconcile its mass, clearances and loads; repeat the matched mission; then build an instrumented coupon or test article when resources and interfaces are available. Gen6 and 1 km/s are future trade targets, not results.

Source trail. P118_gen5_finite_force_map.md; P119_gen5_finite_force_fem2d.md; P120_gen5_finite_coupled_shot.m; P117_rated_energy_mass_audit.md; P116_gen5_assembly_packaging.md; P115_rated_orbit_cartesian.md; GEN5_COMPUTATIONAL_REVIEW.pdf; GEN5_CAD_REVIEW.pdf; CLAIM_EVIDENCE_MAP.md. The complete pack is in the VOLLEY-thesis repository's `university/final_review_2026_10_10/` directory.